30. THE DIAPHRAGM

DURATION : 14:51

STUDY SHEET

COURSE SUMMARY

This video explores the formation of the diaphragm through four essential embryonic structures: the septum transversum, the pleuroperitoneal membranes, the paraxial mesoblast, and the oesophageal mesenchyme. Key concepts include the dynamics of embryonic movement, the functional connection between the diaphragm and the pericardium, and the importance of embryonic flexion for the organisation of internal structures. The interaction between these elements is crucial for cardiac rhythmicity and for the growth of neighbouring organs such as the lungs.

30. THE DIAPHRAGM

The process of **embryonic flexion** is fundamental. It not only organises the establishment of the joints but also initiates a phase of embryonic delimitation. This process involves four main embryonic structures.

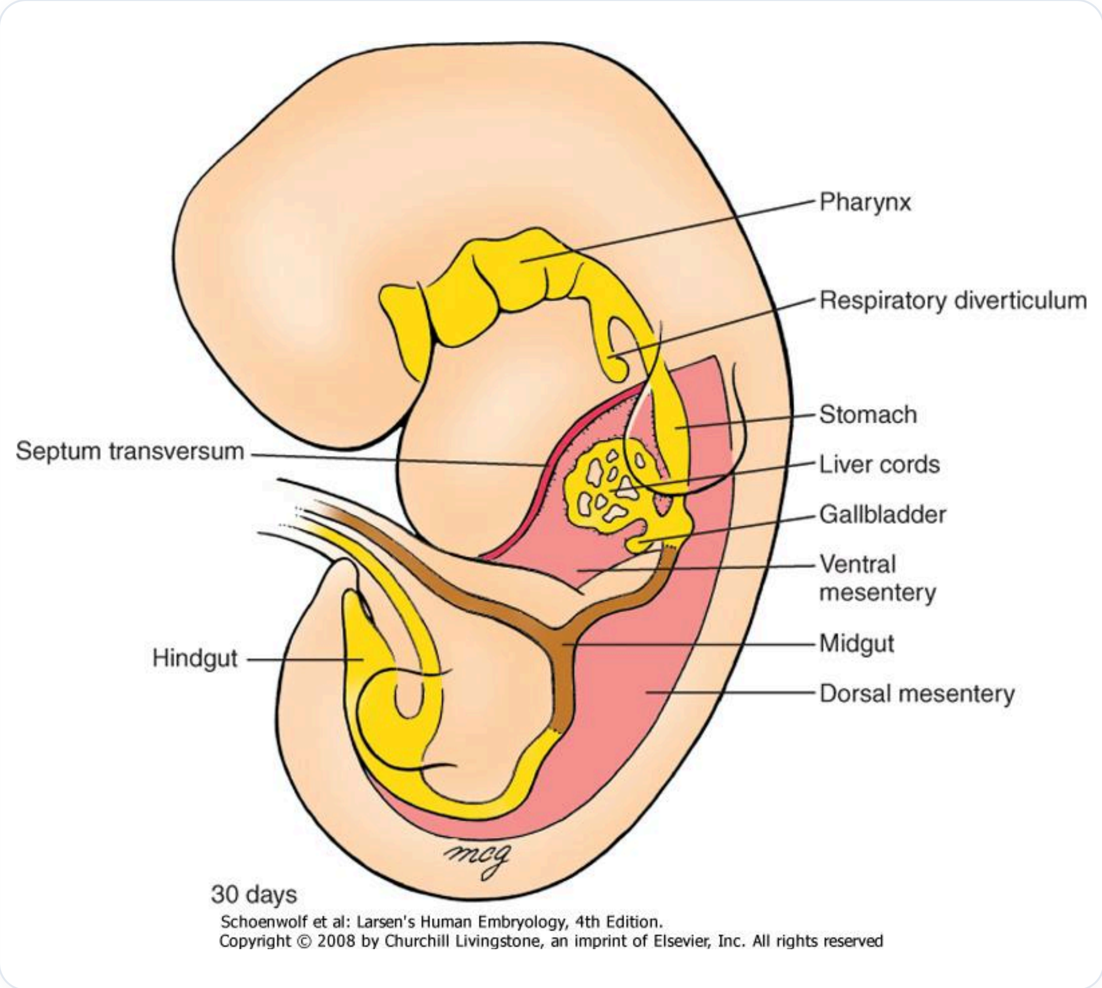

FIG. — Digestive primordia 30 days

THE FOUR EMBRYONIC STRUCTURES OF THE DIAPHRAGM

The four embryonic structures that contribute to the formation of the diaphragm are:

- **The septum transversum**
- **The pleuroperitoneal membranes**
- **The para-axial mesoblast** of the trunk wall
- **The oesophageal mesenchyme**

ORIGIN OF THE PERICARDIUM AND SEPTUM TRANSVERSUM

Beyond the precordial plate, in the cephalic region, the **mesenchymal** cells of the embryonic disc give rise to the pericardium and the septum transversum. It is important to note that your heart is already developing above your head at this stage. This is due to the embryonic flexion movement, which brings these structures into place very early.

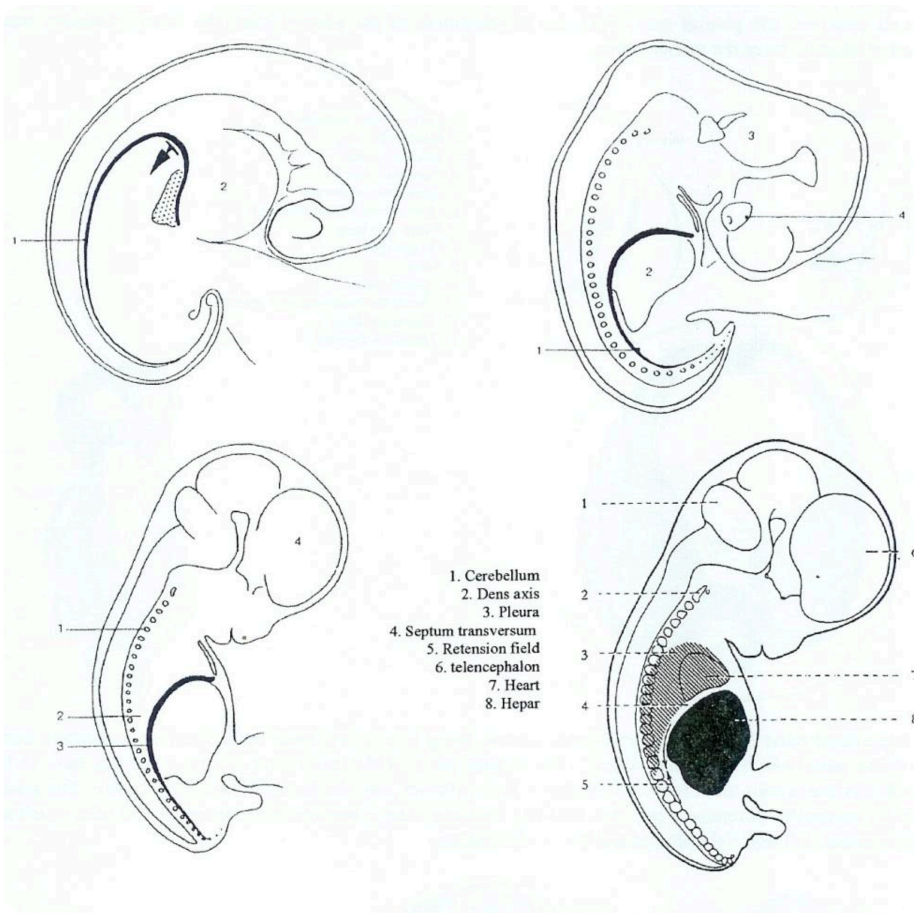

FIG. — Human embryonic development

THE DYNAMICS OF EMBRYONIC MOVEMENT

Contrary to classical embryology, which describes a "descent" of the heart, it is actually the rest of the body that "ascends" around the structures that are already in place. Similarly, for the kidneys, we speak of kidney ascension, but in reality, the kidney remains in place and it is the rest that descends.

THE DIAPHRAGM AND THE PERICARDIUM: A FUNCTIONAL UNIT

The pericardium and the diaphragm form an inseparable **functional unit**. The rhythmicity of the heart is supported by the pericardium. The latter does not merely contain the heart; it actively participates in its function. The closed oscillation system of the pericardium is essential for supporting cardiac rhythmicity.

This functional unit is completely connected between the heart and the diaphragm. They are not two separate entities, but rather structures originating from the same cellular source.

INTEGRATION OF MESENCHYMAL CELLS

The flexion of the embryo, induced by the **notochord**, leads to a specific movement at the level of the neural tube. This neural groove movement organises the lateral structure into small vessels, bringing nutrients but limiting growth. The surrounding ectoderm rolls up, and it is in this movement that mesenchymal cells integrate to form the future pericardial tissue and the diaphragmatic openings.

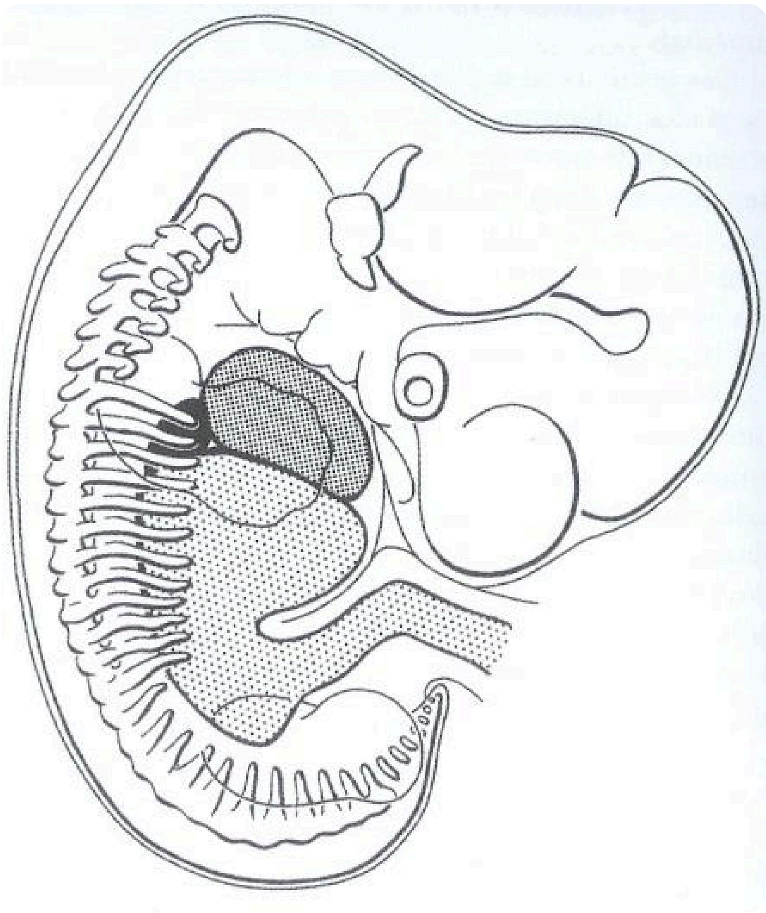

FIG. — Embryo 4 weeks

EVOLUTION OF THE YOLK SAC

As the embryo grows, the **yolk sac** ceases its growth and becomes a mere remnant. It inserts with the vitelline vessels at what is called the umbilical connection. The yolk sac integrates into the pedicle to form the umbilical cord. The pericardium integrates into the septum transversum, whose origin is beyond the precordial plate.

GROWTH OF THE SEPTUM TRANSVERSUM

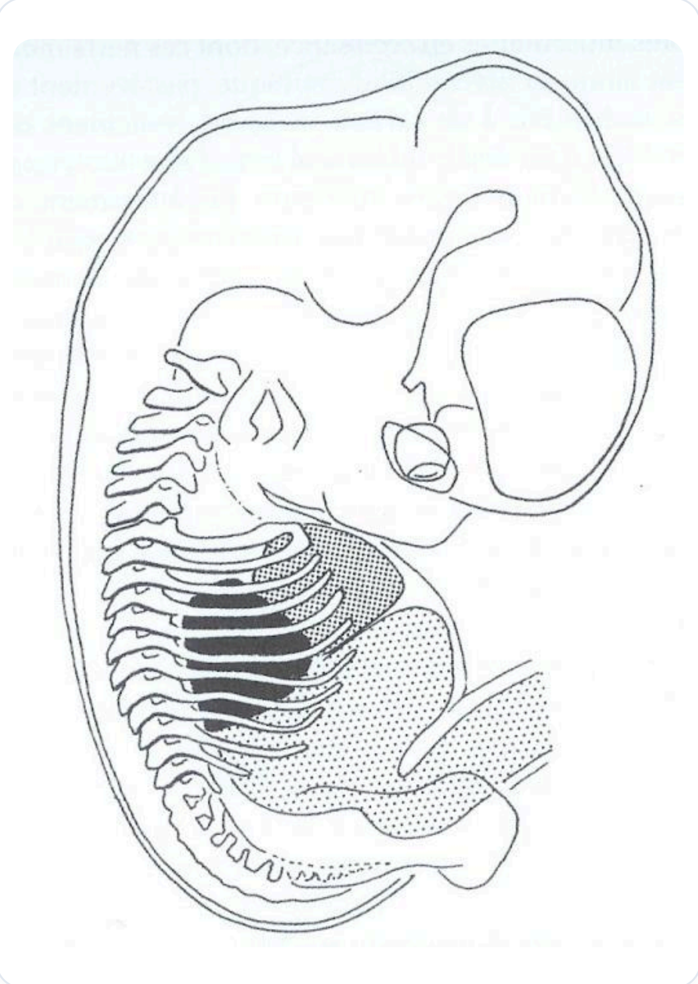

FIG. — Embryo, internal organs

The septum transversum, an important tissue, undergoes significant growth and transformation. In the fourth week, it is initially positioned. In the fifth week, it horizontalises and then begins its "descent". However, this "descent" is actually a consequence of the growth of the rest of the embryo.

THE ORIGIN OF THE PHRENIC NERVE AND THE MOVEMENT OF THE DIAPHRAGM

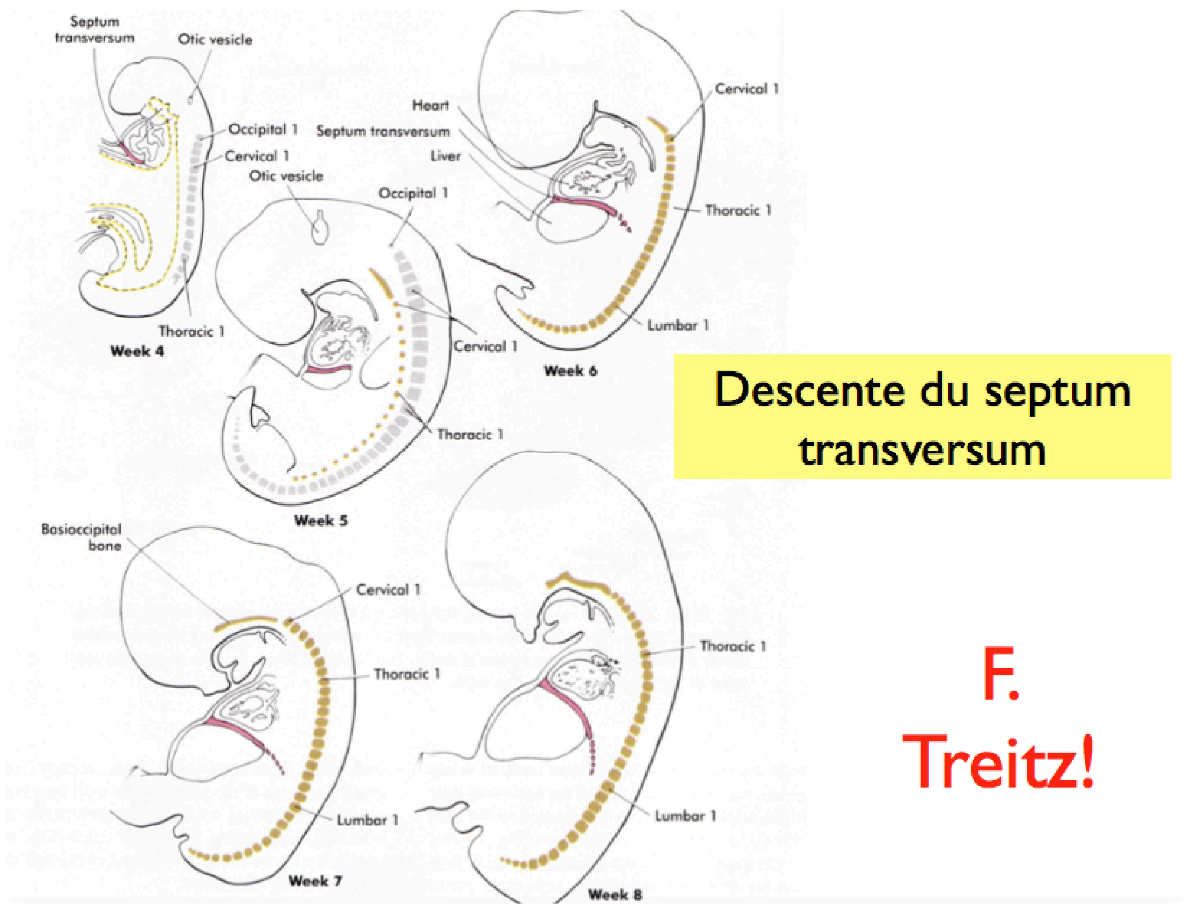

FIG. — Transverse septum descent

The origin of the septum transversum, when the embryo is 2 mm long, is at the very high cervical level (C3, C4), which is precisely the origin of the **phrenic nerve**. The diaphragm eventually ends up at the lumbar level. The idea of "descent" is false in biodynamics. The diaphragm settles, positions itself, and it is the enormous growth of the surrounding structures that modifies it.

CREATION OF SPACES AND ATTRACTION OF THE LUNGS

The growth of the posterior part of the embryo creates an absorption space. This space opens and attracts the lungs, which are a suction field. The lungs are, in a way, drawn into this forming space. The diaphragmatic origin is very high, at the level of the second and third cervical vertebrae.

THE STRAIGHTENING OF THE EMBRYO AND THE POSITIONING OF THE DIAPHRAGM

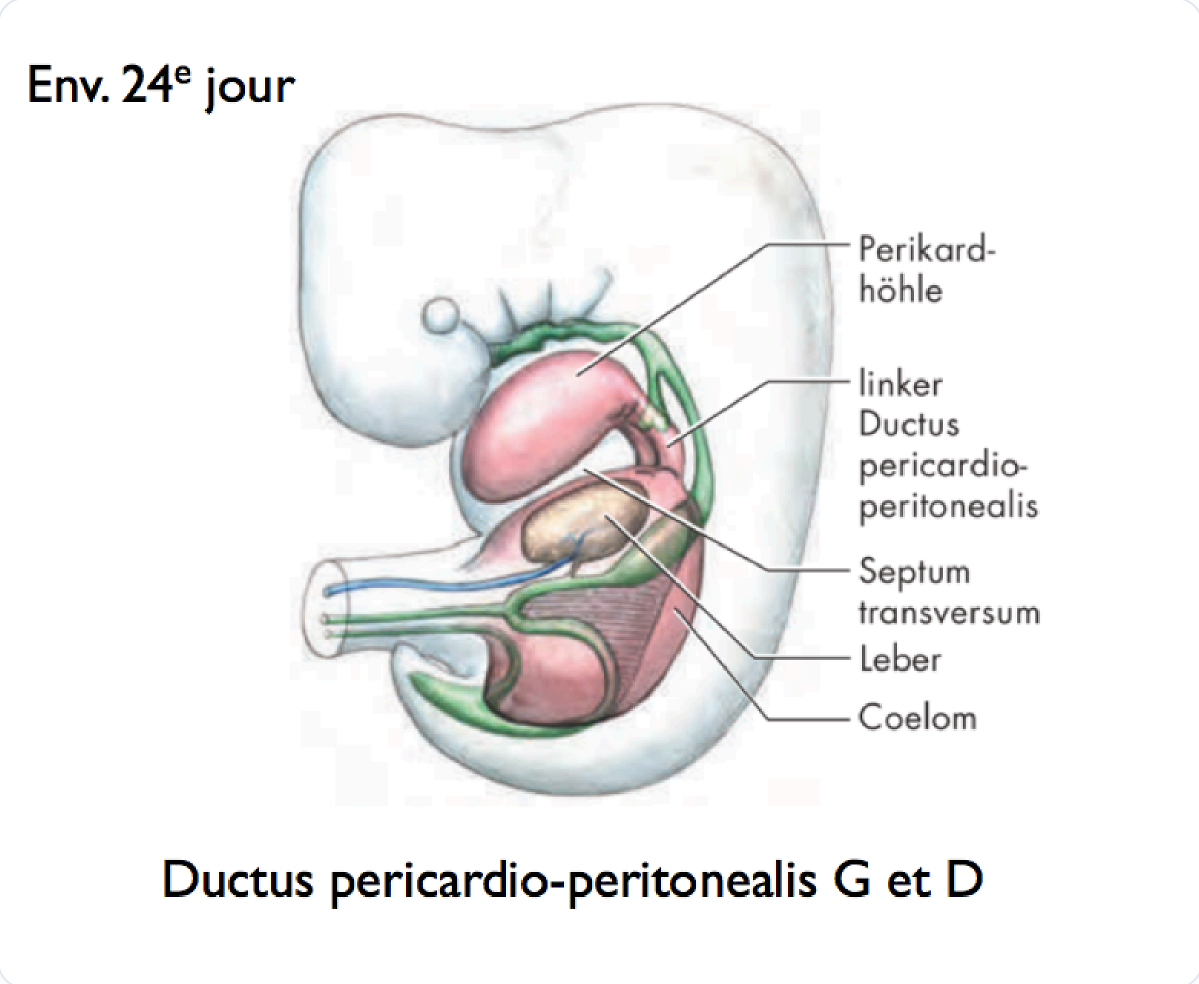

FIG. — Ductus pericardio-peritonealis

The phrenic nerve is not "pulled" but rather "accompanied" by this growth. The positioning of the diaphragm is linked to excessive growth. At a certain point, the embryo stops flexing and straightens up. This straightening is a matter of fluids on the lateral sclerotome, with the ascent of the telencephalon, the neural tube, and the descent of the digestive system. This movement is crucial for integration and the creation of spaces.

THE DIFFERENT TISSUES OF THE DIAPHRAGM

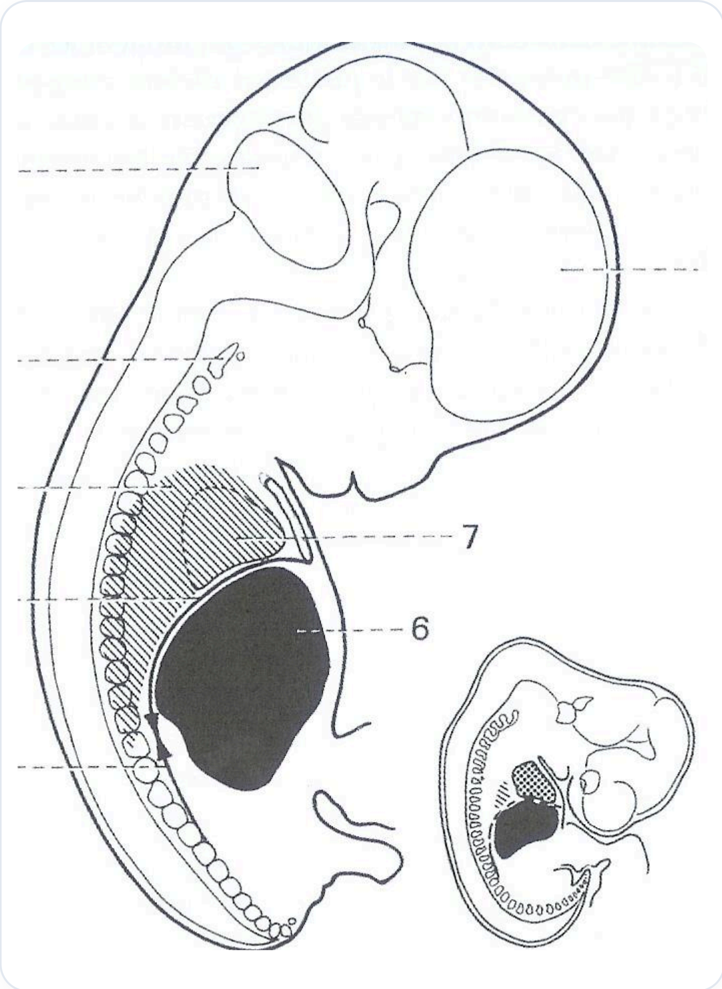

FIG. — Human embryo 6 weeks

On a section, one can observe the different tissues of the diaphragm:

- **Septum transversum**
- **Pleuroperitoneal membranes**
- **Oesophageal mesentery**
- **Para-axial membranes**

Working on a diaphragm is not limited to these spaces, but also to the release of the aorta, the vena cava, the oesophagus, and especially the **oesophageal junction**.

THE IMPORTANCE OF THE OESOPHAGEAL JUNCTION

The oesophageal junction is a very important area for releasing a diaphragm. Oesophageal and peptic disorders are common and often linked to a blockage in this area. It is essential to release the peri-oesophageal space and associated muscles. This small space can block the entire digestive system, suspended from the carotid foramina, the pre-jugular system, and the pterygoid attachments at the base of the skull.

FUNCTIONAL CONTINUITY AND THE LIGAMENT OF TREITZ

There is a functional continuity with a small muscle, the **ligament of Treitz** (or suspensory muscle of the duodenum), which is vital for the opening of sphincters and the organisation of the metabolic and digestive system. This muscle contains two types of fibres, muscular and non-muscular, offering different information. It can open or close the air zones.

CONNECTIONS AND DERIVATIONS OF THE SEPTUM TRANSVERSUM

The **falciform ligament** and the **greater omentum** are elements derived from the septum transversum. There is therefore a continuity from the mesenchymal cells, via the pericardial system and the septum transversum, and from the latter derive the falciform ligament and the greater omentum.

The greater omentum is an integral part of diaphragmatic function. It ascends towards the bronchial area, orientates towards the spleen, then descends above the colon to form a coalescence of four layers, creating a large **mesangiolar** omentum (lymphatic B, T and M). Its motility is related to the diaphragmatic system.

RELATIONS WITH THE ARCUATE LIGAMENTS

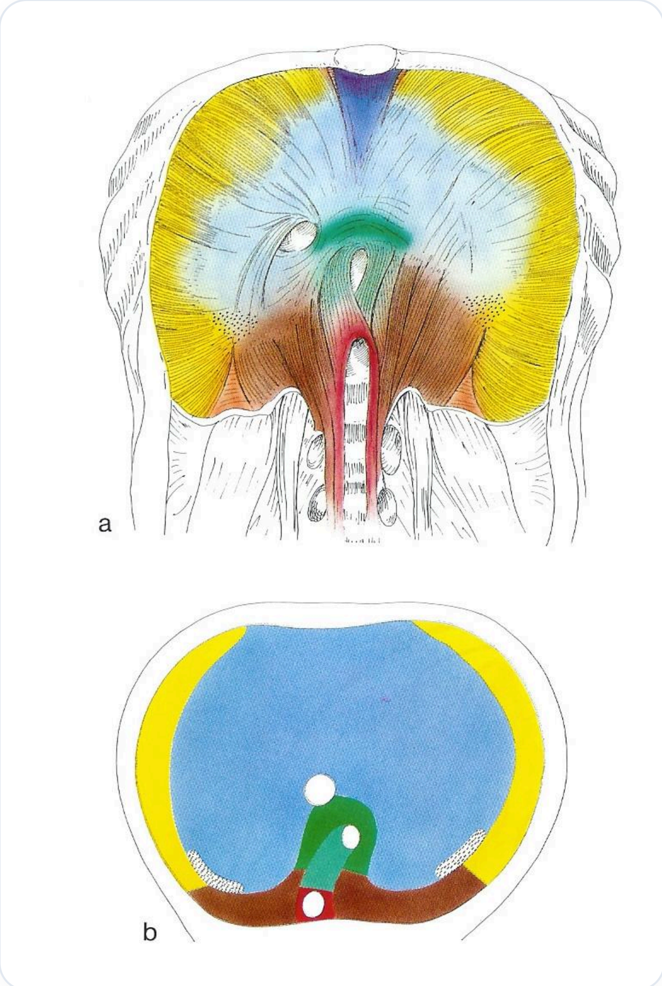

FIG. — Diaphragm skeletal muscle

The **arcuate ligaments** play a crucial role in the diaphragm's relationship with surrounding structures:

- The medial arcuate ligament attaches to the psoas arch.
- The lateral arcuate ligament attaches to the quadratus lumborum.
- The median arcuate ligament attaches to the aortic hiatus.

This continuity extends to the ligament of Treitz. The diaphragm continues indirectly through the common masses, the quadratus lumborum, and the psoas muscles. This is a fascial continuity of diaphragmatic organisation.

THE DIAPHRAGM: AN ESSENTIAL REGULATOR

An oesophageal issue can disrupt the movement of a knee, for example, by affecting overall mechanics. Each diaphragm must be considered in relation to the others. The three main diaphragms are:

1. **The cervical diaphragm**

2. **The thoracic diaphragm**

3. **The pelvic diaphragm**

An imbalance in one disrupts the other. The thoracic diaphragm is a pressure, tension, hydraulic, and thermal regulator.

MULTIPLE FUNCTIONS OF THE DIAPHRAGM

The diaphragm is a **lymphatic pump**, essential for the digestive, vascular, and lymphatic systems. Its relationship with the oesophagus, the cardiac system (especially the cardiac relationship with "phoepactis"), and the greater omentum is fundamental. Posteriorly, it is related to the quadratus lumborum and the psoas muscles. This continuity extends from the heart to the oesophagus, up to the base of the skull, in connection with the craniosacral axis.

THE COSTODIAPHRAGMATIC RECESSES

The **costodiaphragmatic recesses** are accumulation sacs for exudates and waste, whether transperitoneal or transpleural. During an infection such as bronchitis, microbes and mucus are eliminated and can accumulate in these sacs. This is why, after an infection, the lung bases can be blocked by exudates. These accumulations require space to be managed by the body.

THE CELLULAR AND TISSUE UNITY OF THE BODY

The human body is a **cellular and tissue unit**. Its defence and integration reactions can manifest at any level thanks to the tissue continuity of the splanchnopleure and somatopleure, as we have seen with the diaphragm.

FIG. — Thoracic abdominal sagittal section